STEP 1 Install required libraries

```
!pip install spacy pandas matplotlib seaborn emoji
!python -m spacy download en_core_web_sm
```

```
Requirement already satisfied: spacy in /usr/local/lib/python3.12/dist-packages (3.8.11)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)
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Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests<3.0.0,>=2.13.0->spacy) (3.11)
```

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Installing collected packages: emoji
Successfully installed emoji-2.15.0
Collecting en-core-web-sm==3.8.0
  Downloading https://github.com/explosion/spacy-models/releases/download/en_core_web_sm-3.8.0/en_core_web_sm-3.8.0-py3-none-any.whl
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✓ Download and installation successful
```

STEP 2: Import libraries

```
import pandas as pd
import re
import spacy
import emoji
from collections import Counter
import matplotlib.pyplot as plt
import seaborn as sns
```

STEP 3: Load the dataset

```
df = pd.read_csv('/content/Tweets.csv')
df.head()
```

	tweet_id	airline_sentiment	airline_sentiment_confidence	negativereason	negativereason_confidence	airline	airline_se
0	570306133677760513	neutral	1.0000	NaN	NaN	Virgin America	
1	570301130888122368	positive	0.3486	NaN	0.0000	Virgin America	
2	570301083672813571	neutral	0.6837	NaN	NaN	Virgin America	
3	570301031407624196	negative	1.0000	Bad Flight	0.7033	Virgin America	
4	570300817074462722	negative	1.0000	Can't Tell	1.0000	Virgin America	

```
df = pd.read_csv("Tweets.csv")
df.head()
```

	tweet_id	airline_sentiment	airline_sentiment_confidence	negativereason	negativereason_confidence	airline	airline_se
0	570306133677760513	neutral	1.0000	NaN	NaN	Virgin America	
1	570301130888122368	positive	0.3486	NaN	0.0000	Virgin America	
2	570301083672813571	neutral	0.6837	NaN	NaN	Virgin America	
3	570301031407624196	negative	1.0000	Bad Flight	0.7033	Virgin America	
4	570300817074462722	negative	1.0000	Can't Tell	1.0000	Virgin America	

STEP 4: Explore the dataset

```
print("Dataset Shape:", df.shape)
print("\nColumns:\n", df.columns)

# Sentiment distribution
df['airline_sentiment'].value_counts()
```

```
Dataset Shape: (14640, 15)

Columns:
 Index(['tweet_id', 'airline_sentiment', 'airline_sentiment_confidence',
       'negativereason', 'negativereason_confidence', 'airline',
       'airline_sentiment_gold', 'name', 'negativereason_gold',
       'retweet_count', 'text', 'tweet_coord', 'tweet_created',
       'tweet_location', 'user_timezone'],
      dtype='object')
```

airline_sentiment	count
negative	9178
neutral	3099
positive	2363

dtype: int64

STEP 5: Define tweet cleaning function

```
def clean_tweet(text):
    text = re.sub(r"http\S+|www\S+", "", text)      # remove URLs
    text = re.sub(r"@\w+", "", text)                # remove mentions
    text = emoji.replace_emoji(text, replace='')   # remove emojis
    text = re.sub(r"[^a-zA-Z# ]", "", text)         # keep hashtags
    text = re.sub(r"\s+", " ", text).strip()        # remove extra spaces
    return text.lower()
```

STEP 6: Apply cleaning

```
df['cleaned_text'] = df['text'].apply(clean_tweet)
df[['text', 'cleaned_text']].head()
```

	text	cleaned_text
0	@VirginAmerica What @dhepburn said.	what said
1	@VirginAmerica plus you've added commercials t...	plus youve added commercials to the experience...
2	@VirginAmerica I didn't today... Must mean I n...	i didnt today must mean i need to take another...
3	@VirginAmerica it's really aggressive to blast...	its really aggressive to blast obnoxious enter...
4	@VirginAmerica and it's a really big bad thing...	and its a really big bad thing about it

STEP 7: Load spaCy model

```
nlp = spacy.load("en_core_web_sm")
```

STEP 8: Process tweets with spaCy (Lemmas & POS)

```
lemmas_list = []
pos_list = []

for doc in nlp.pipe(df['cleaned_text']):
    lemmas = [token.lemma_ for token in doc if not token.is_stop]
    pos = [token.pos_ for token in doc if not token.is_stop]

    lemmas_list.append(lemmas)
    pos_list.append(pos)

df['lemmas'] = lemmas_list
df['pos_tags'] = pos_list

df[['cleaned_text', 'lemmas', 'pos_tags']].head()
```

	cleaned_text	lemmas	pos_tags
0	what said	[say]	[VERB]
1	plus youve added commercials to the experience...	[plus, ve, add, commercial, experience, tacky]	[CCONJ, AUX, VERB, NOUN, NOUN, ADV]
2	i didnt today must mean i need to take another...	[not, today, mean, need, trip]	[PART, NOUN, VERB, VERB, NOUN]
3	its really aggressive to blast obnoxious enter...	[aggressive, blast, obnoxious, entertainment, ...	[ADJ, VERB, ADJ, NOUN, NOUN, VERB, ADJ, ADJ, N...

STEP 9: Create custom spaCy pipeline component (Hashtag Detector)

```
from spacy.language import Language

@Language.component("hashtag_detector")
def hashtag_detector(doc):
    hashtags = [token.text for token in doc if token.text.startswith("#")]
    doc._.hashtags = hashtags
    return doc
```

STEP 10: Register custom attribute and add pipeline

```
from spacy.tokens import Doc

Doc.set_extension("hashtags", default=[])

nlp.add_pipe("hashtag_detector", last=True)

nlp.pipe_names
```

```
['tok2vec',
 'tagger',
 'parser',
 'attribute_ruler',
 'lemmatizer',
 'ner',
 'hashtag_detector']
```

STEP 11: Extract hashtags

```
all_hashtags = []

for doc in nlp.pipe(df['cleaned_text']):
    all_hashtags.extend(doc._.hashtags)

hashtag_freq = Counter(all_hashtags)
hashtag_freq.most_common(10)
```

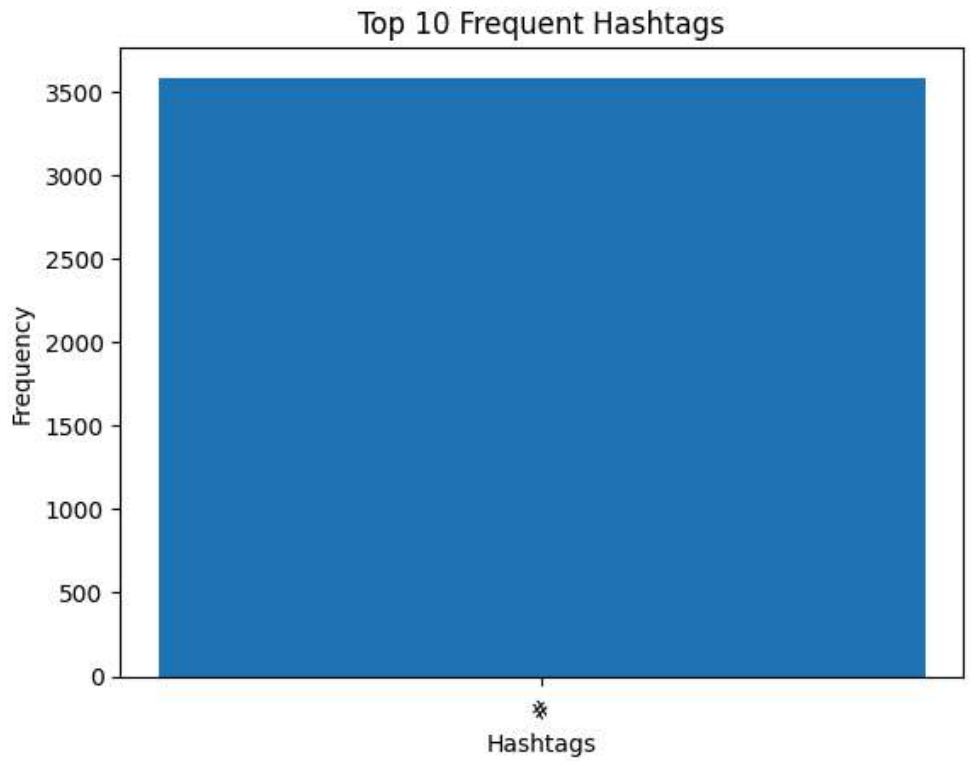
```
[('#', 3583)]
```

STEP 12: Visualize frequent hashtags

```
top_hashtags = hashtag_freq.most_common(10)

hashtags, counts = zip(*top_hashtags)

plt.figure()
plt.bar(hashtags, counts)
plt.xticks(rotation=45)
plt.title("Top 10 Frequent Hashtags")
plt.xlabel("Hashtags")
plt.ylabel("Frequency")
plt.show()
```



STEP 13: Filter negative tweets

```
negative_tweets = df[df['airline_sentiment'] == 'negative']
negative_tweets.shape
```

```
(9178, 18)
```

STEP 14: POS distribution in negative tweets

```
neg_pos = []

for doc in nlp.pipe(negative_tweets['cleaned_text']):
    neg_pos.extend([token.pos_ for token in doc if not token.is_stop])

pos_freq = Counter(neg_pos)
```

```
Counter({'ADJ': 8171,
         'VERB': 22687,
         'NOUN': 37090,
         'ADV': 2625
```